

# Cassava Leaf Disease Classification Using Lightweight Deep Learning Model

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- World population to reach 9.1 billion by 2050, which requires 70% growth in food production. [15]
- Key Factors affecting plants and its products
  - **Diseases:** Biotic factors caused by fungi, bacteria or algae.
  - **Disorders:** Environment related, Nutrient deficiency etc.
- Plant diseases:
  - Cause severe threat to global food security.
  - Accounts for 20%-40% of all crop losses globally. [10]

# Introduction (Contd.)

**Table 1:** Yield loss by Various Types of Plant Diseases.

| <b>Plant Part</b>    | <b>Im-</b> | <b>Type of Disease</b>         | <b>Percentage Loss</b>  | <b>Yield</b> |
|----------------------|------------|--------------------------------|-------------------------|--------------|
| Leaves (Foliar)      |            | Leaf spots, rusts, blights     | 24% – 41% [4]           |              |
| Roots                |            | Root rots, wilts, nematodes    | Variable, up to 20% [4] |              |
| Stems/Collar         |            | Cankers, stem rots             | 5% – 15% [4, 10]        |              |
| Fruits/Flowers       |            | Fruit rots, scab, petal blight | 10% – 20% [4]           |              |
| Whole Plant/Systemic |            | Viral wilts, systemic blights  | Up to 15% [4]           |              |

# Introduction (Contd.)

## Leaf Diseases

- Strong impact on yield due to reduced photosynthesis.
- Plant Leaf Diseases have Distinctive characteristics:
  - Randomly Distributed Lesions.
  - Diverse Symptoms.
  - Complex Backgrounds.

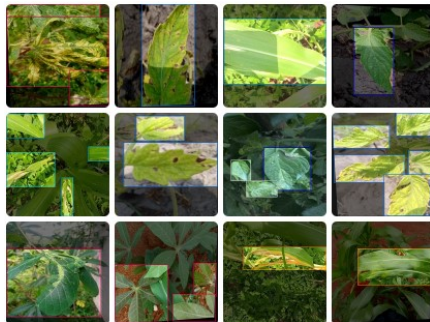


Figure 1: Plant Leaf Disease Sample Images

## Plant Leaf Disease Diagnosis

- **Traditional Method:**

- Molecular Technology Based: Complicated and Time consuming, Constrained to Centralized Laboratories.

- **Image Processing Based Methods:**

- Hyper spectral Image Analysis, Image Segmentation.

- **Machine Learning and Deep Learning Based Methods:**

- Automated, Accurate and Fast.

# Literature Survey I

| Publication                          | Title                                                                                                       | Methodology Used                                                                                                                                                                          | Characteristics                                                                                                                                                                             | Limitations                                                                                                                                                                             |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Expert Syst. Appl. (2025) [7]        | An Enhanced Vision Transformer Network for Efficient and Accurate Crop Disease Detection                    | <ul style="list-style-type: none"> <li>Improved ViT with Triplet Multi-Head Attention (t-MHA)</li> <li>Transfer learning comparison</li> <li>LIME &amp; t-SNE explainability</li> </ul>   | <ul style="list-style-type: none"> <li>97.99% acc. on RiceCApp, 98.57% on PlantVillage</li> <li>Lightweight: ~2.25M params</li> <li>Field RGB images (Rice &amp; Apple)</li> </ul>          | <ul style="list-style-type: none"> <li>Only 2 crops tested (Rice &amp; Apple)</li> <li>Noisy background from field images</li> <li>Hard to scale to more crop species</li> </ul>        |
| Comput. Electron. Agric. (2022) [9]  | P2OP — Plant Pathology on Palms: A Deep Learning-Based Mobile Solution for In-Field Plant Disease Detection | <ul style="list-style-type: none"> <li>Deep Metric Learning (Siamese Network)</li> <li>Simplified MobileNetV2 embedding</li> <li>Constrained K-Means prototype clustering</li> </ul>      | <ul style="list-style-type: none"> <li>97%, 97.1%, 96.4% acc. (Apple, Citrus, Tomato)</li> <li>Only 0.26M parameters, 1MB storage</li> <li>Offline Android smartphone deployment</li> </ul> | <ul style="list-style-type: none"> <li>Requires manual leaf cropping by user</li> <li>Trained on small per-class sets (100 img)</li> <li>Cannot handle unknown/new diseases</li> </ul>  |
| Comput. Electron. Agric. (2023) [17] | FormerLeaf: An Efficient Vision Transformer for Cassava Leaf Disease Detection                              | <ul style="list-style-type: none"> <li>Vision Transformer (ViT) backbone</li> <li>LeIAP attention head pruning</li> <li>Sparse Matrix-Matrix Multiplication (SPMM)</li> </ul>             | <ul style="list-style-type: none"> <li>96.82% F1 on Cassava dataset</li> <li>Model size reduced 28% after pruning</li> <li>Training time -10%, inference -15%</li> </ul>                    | <ul style="list-style-type: none"> <li>Tested only on Cassava leaf dataset</li> <li>Dataset heavily imbalanced (60% CMD)</li> <li>High param count vs. CNN counterparts</li> </ul>      |
| Expert Syst. Appl. (2025) [8]        | An Enhanced Vision Transformer for Efficient and Accurate Crop Disease Detection (RicApp & PlantVillage)    | <ul style="list-style-type: none"> <li>Enhanced ViT with t-MHA (3 cascaded units)</li> <li>Data augmentation (rotation, flip, gamma)</li> <li>Ablation on encoder/patch setups</li> </ul> | <ul style="list-style-type: none"> <li>Outperforms 12 SOTA CNN &amp; ViT models</li> <li>Statistical validation (paired t-test)</li> <li>LIME &amp; t-SNE explainability</li> </ul>         | <ul style="list-style-type: none"> <li>Complex background slightly lowers acc.</li> <li>Misclassifies similar symptom classes</li> <li>Needs high compute (NVIDIA A100 GPUs)</li> </ul> |

# Literature Survey II

| Publication                                                  | Title                                                                            | Methodology Used                                                                                                                                                                                                                               | Characteristics                                                                                                                                                                                                                          | Limitations                                                                                                                                                                                                                            |
|--------------------------------------------------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Thakur et al. (2023)<br>Ecological Informatics [19]          | Vision Transformer Meets CNN for Plant Disease Classification                    | <ul style="list-style-type: none"> <li>Hybrid CNN+ViT (VGG16 + Inception v7 + 4 Transformer blocks)</li> <li>Grad-CAM &amp; LIME explainability</li> <li>5 public datasets</li> </ul>                                                          | <ul style="list-style-type: none"> <li>98.86% on PlantVillage, 89.24% on Embrapa</li> <li>Only 0.85M parameters</li> <li>Suitable for IoT-based smart agriculture</li> </ul>                                                             | <ul style="list-style-type: none"> <li>High FLOPS count limits IoT deployment</li> <li>Multi-disease leaf samples cause misclassification</li> <li>Struggles with highly imbalanced datasets</li> </ul>                                |
| Bhagat et al. (2024)<br>Smart Agricultural Technology [3]    | Advancing Real-Time Plant Disease Detection: Lite-MDC & Novel Pigeon Pea Dataset | <ul style="list-style-type: none"> <li>Lite-MDC: Multi-kernel Depthwise Separable CNN (MDsConv)</li> <li>VGG16-inspired backbone with <math>3 \times 3</math> &amp; <math>5 \times 5</math> kernels</li> <li>Grad-CAM visualization</li> </ul> | <ul style="list-style-type: none"> <li>94.14% on novel pigeon pea dataset; 99.78% PlantVillage</li> <li>Only 2.2M params, 34 FPS on NVIDIA P100</li> <li>First real-field pigeon pea dataset (18 Indian states)</li> </ul>               | <ul style="list-style-type: none"> <li>Classification only — no disease localization/bounding box</li> <li>Performance drops under heavy background noise</li> <li>Needs object detection for precise region identification</li> </ul> |
| Nawaz et al. (2024)<br>Expert Systems With Applications [12] | CoffeeNet: A Deep Learning Approach for Coffee Plant Leaves Diseases Recognition | <ul style="list-style-type: none"> <li>CenterNet + CBAM attention-based ResNet-50</li> <li>Spatial-channel attention (CBAM) feature extractor</li> <li>1-stage object detection with heatmap/offset/dimension heads</li> </ul>                 | <ul style="list-style-type: none"> <li>98.54% accuracy, mAP 0.97 on Arabica coffee leaf dataset</li> <li>Handles 5 disease classes under complex real-world conditions</li> <li>Outperforms FastRCNN, YOLO-V3, SSD, RetinaNet</li> </ul> | <ul style="list-style-type: none"> <li>Trained on single-crop coffee dataset only</li> <li>High similarity between healthy and Miner class causes errors</li> <li>Needs larger, more diverse datasets for generalization</li> </ul>    |



# Literature Survey III

| Publication                          | Title                                                                                                                 | Methodology Used                                                                                                                                                                                                                                                                    | Characteristics                                                                                                                                                                                                                | Limitations                                                                                                                                                                                                                               |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bakr et al. (2022) EJECE Journal [2] | DenseNet Based Model for Plant Diseases Diagnosis                                                                     | <ul style="list-style-type: none"> <li>Transfer learning: DenseNet201 as feature extractor + CNN classifier</li> <li>Compared VGG16, InceptionV3, ResNet152V2, DenseNet201</li> <li>Web-based Flask application for farmer diagnosis</li> </ul>                                     | <ul style="list-style-type: none"> <li>99.44% training, 98.70% validation accuracy</li> <li>15-class detection (Tomato, Potato, Pepper diseases)</li> <li>Web app provides disease info + treatment recommendations</li> </ul> | <ul style="list-style-type: none"> <li>Limited to 3 crops only (tomato, potato, pepper)</li> <li>Dataset from controlled PlantVillage — not real-field images</li> <li>No evaluation on in-field or out-of-distribution images</li> </ul> |
| Bajpai & Sahu (2026) [1]             | Lightweight and Interpretable Hybrid Deep Network for Potato Leaf Disease Detection Using CBAM and Vision Transformer | <ul style="list-style-type: none"> <li>Ghost Conv for lightweight features</li> <li>CBAM (spatial &amp; channel attention)</li> <li>LiteViT blocks for global dependencies</li> <li>5-fold CV &amp; mixed precision training</li> </ul>                                             | <ul style="list-style-type: none"> <li>99.67% test acc. on field potato dataset</li> <li>~4.7M params, lightweight</li> <li>Interpretable via CBAM attention maps</li> <li>Outperforms ensemble CNN (98.49%)</li> </ul>        | <ul style="list-style-type: none"> <li>Only potato (Early/Late Blight, Healthy)</li> <li>Single-location field dataset (Kannauj, India)</li> <li>Scaling to more crops &amp; edge devices needed</li> </ul>                               |
| Singla et al. (2024) [16]            | Detection of plant leaf diseases using deep convolutional neural network models                                       | <ul style="list-style-type: none"> <li>Proposed 12-layer CNN (4 Conv + 4 MaxPool + Dense)</li> <li>Compared MobileNet, DenseNet201, ResNet50, InceptionV3, VGG16/19</li> <li>Binary &amp; multiclass (38 classes) classification</li> <li>Flask web app with email alert</li> </ul> | <ul style="list-style-type: none"> <li>99.39% binary acc. (scratch model)</li> <li>MobileNet: 97.35% multiclass acc.</li> <li>Web-based app for farmers</li> <li>Comprehensive comparative study</li> </ul>                    | <ul style="list-style-type: none"> <li>PlantVillage dataset only (lab-controlled)</li> <li>No real-field testing</li> <li>Multiclass limited to lab-setting crops</li> </ul>                                                              |
| Singh et al. (2020) [14]             | PlantDoc: A Dataset for Visual Plant Disease Detection                                                                | <ul style="list-style-type: none"> <li>Curated PlantDoc dataset (2,598 images, 13 species, 27 classes) from web</li> <li>Benchmarked VGG16, InceptionV3, InceptionResNetV2, Faster R-CNN</li> <li>Transfer learning experiments</li> </ul>                                          | <ul style="list-style-type: none"> <li>Real-world plant disease detection dataset</li> <li>Fine-tuning on PlantDoc improved acc. up to 31%</li> <li>Publicly available dataset &amp; code</li> </ul>                           | <ul style="list-style-type: none"> <li>Small dataset (2,598 images)</li> <li>Class imbalance, web-sourced noise</li> <li>No mobile deployment demonstrated</li> </ul>                                                                     |

# Literature Survey IV

| Publication                  | Title                                                                      | Methodology Used                                                                                                                                                                                                                        | Characteristics                                                                                                                                                                                                               | Limitations                                                                                                                                                                       |
|------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Usha Ruby et al. (2024) [13] | Wheat leaf disease classification using a modified ResNet50 CNN model      | <ul style="list-style-type: none"> <li>Modified ResNet50 (added Conv, BN, LeakyReLU)</li> <li>Collaborative GAN for missing data imputation</li> <li>Data augmentation &amp; transfer learning</li> </ul>                               | <ul style="list-style-type: none"> <li>98.44% acc. on LWDD2020 wheat dataset</li> <li>Outperforms ResNet50, InceptionV3, DenseNet</li> <li>4-class (Leaf Rust, Crown/Root Rot, Loose Smut, Healthy)</li> </ul>                | <ul style="list-style-type: none"> <li>Tested only on wheat</li> <li>Limited disease classes</li> <li>May not generalise to other crops/field conditions</li> </ul>               |
| Chen et al. (2021) [5]       | Identification of rice plant diseases using lightweight attention networks | <ul style="list-style-type: none"> <li>MobileNet-V2 backbone + CBAM (channel &amp; spatial attention)</li> <li>Twice transfer learning &amp; optimised loss</li> <li>Data augmentation (traditional + DCGAN)</li> </ul>                 | <ul style="list-style-type: none"> <li>99.67% acc. on PlantVillage (10 classes)</li> <li>98.48% on local rice dataset with complex backgrounds</li> <li>Lightweight (MobileNet-V2 based)</li> </ul>                           | <ul style="list-style-type: none"> <li>Small local dataset (1,100 images)</li> <li>Limited to rice diseases</li> <li>Complex background still challenging</li> </ul>              |
| Thakur et al. (2023) [18]    | VGG-ICNN: A Lightweight CNN Model for Crop Disease Identification          | <ul style="list-style-type: none"> <li>Hybrid VGG16 (first 4 layers) + 3 Inception v7 blocks</li> <li>7 Conv layers, Global Average Pooling</li> <li>Transfer learning from ImageNet</li> <li>Evaluated on 5 public datasets</li> </ul> | <ul style="list-style-type: none"> <li>99.16% PlantVillage, 93.66% Embrapa, 94.24% Apple, 91.36% Maize, 96.67% Rice</li> <li>~6M params, lightweight</li> <li>Consistent performance across diverse crops/datasets</li> </ul> | <ul style="list-style-type: none"> <li>FLOPs (45.7M) higher than ultra-lightweight</li> <li>May limit real-time on low-end devices</li> <li>Not tested on edge devices</li> </ul> |

- **Performance Drop in Field Conditions:** Models trained on lab images often underperform on real-field data due to environmental variability.
- **Complex Background Interference:** Lightweight deep learning models often show limited performance when cassava leaf images contain complex backgrounds (such as soil, weeds, overlapping leaves, or varying lighting conditions). This drop in accuracy under real-world, unconstrained environments remains a key challenge that is not yet fully addressed.

# Objectives

- ① To design deep learning models that effectively addresses the challenges of plant leaf disease classification under real-world field conditions.
- ② To develop a lightweight preprocessing pipeline that effectively suppresses complex backgrounds and enhances diseased region focus.

# Dataset Used

## FieldPlant:

- Contains 10,326 real-world images of healthy and diseased plant leaves captured under diverse field conditions.
- Downloaded it from the Roboflow in the YOLO format.
- Covers 5 different plant species and 27 disease classes.



Figure 2: FieldPlant: Sample Images

# Dataset Used (Contd.)

**Cassava Leaf Disease Classification:** This dataset contains the total 21,397 images which belong to the five different classes

- Cassava Bacterial Blight (CBB): 1087
- Cassava Brown Streak Disease (CBSD): 2189
- Cassava Green Mottle (CGM): 2386
- Cassava Mosaic Disease (CMD): 13158
- Healthy: 2577

The image was captured in real life field.



Figure 3: Cassava: Sample Images

# Model Pipeline

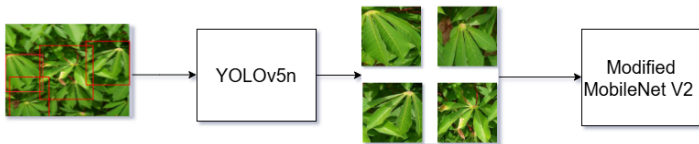


Figure 4: Model Pipeline

# Stage 1: Leaf Extraction

## Plant Leaves Extraction:

- Plant's leaves are detected and extracted from image
- Removes the impact of complex background while training the model.

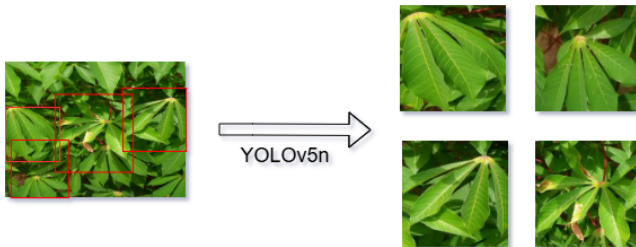


Figure 5: Stage-1: Plants Leaves Extraction



# Stage 2: Classification

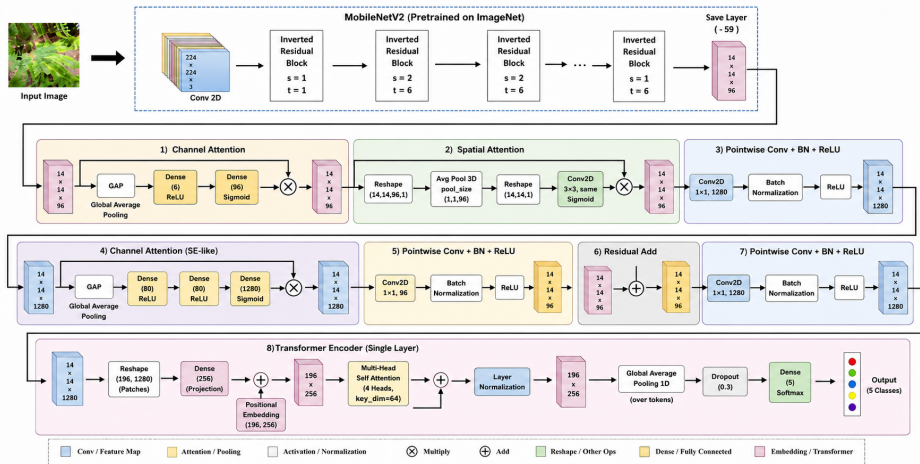


Figure 6: Proposed architecture for classification

# Result and Analysis

| Model Name                    | Trainable Parameters | Non Trainable Parameter | Model Size     |
|-------------------------------|----------------------|-------------------------|----------------|
| MobileNet V2                  | 2,230,277            | 34,112                  | 8.51 MB        |
| MobileNet V3-S                | 2,547,861            | 12,112                  | 9.72 MB        |
| Shufflenet v2                 | 1,509,209            | 7,114                   | 7.65 MB        |
| EfficientNet V2-S             | 21,463,493           | 153,872                 | 81.88 MB       |
| LMFRNet                       | 1,518,445            | 3,680                   | 6.98 MB        |
| EffecientNet.Lite             | 3,377,413            | 42,016                  | 12.88 MB       |
| EfficientNet.Lite + Attention | 1,655,381            | 6,304                   | 7.23 MB        |
| RiceDisease [6]               | 1,854,904            | 2,260,544               | 7.08 MB        |
| CoffeeNet [11]                | 26,139,776           | 54,400                  | 99.72 MB       |
| <b>Proposed Model</b>         | <b>1,489,397</b>     | <b>16,448</b>           | <b>5.68 MB</b> |

**Table 2:** The actual memory and the number of parametr s used by different SOTA and lightweight CNN models along with the proposed model

# Result and Analysis

| Model Name                    | Precision  | Recall     | F1-Score   | Accuracy   |
|-------------------------------|------------|------------|------------|------------|
| MobileNet V2                  | 42%        | 42%        | 41%        | 42%        |
| MobileNet V3                  | 66%        | 62%        | 61%        | 62%        |
| Shufflenet v2                 | 38%        | 36%        | 38%        | 47%        |
| EfficientNet V2-S             | 57%        | 56%        | 56%        | 56%        |
| LMFRNet                       | 70%        | 69%        | 69%        | 69%        |
| EfficientNet.Lite             | 71%        | 71%        | 71%        | 71%        |
| EfficientNet.Lite + Attention | 70%        | 70%        | 68%        | 70%        |
| RiceDisease [6]               | 49%        | 46%        | 46%        | 46%        |
| CoffeeNet [11]                | 29%        | 28%        | 28%        | 28%        |
| <b>Proposed Model</b>         | <b>73%</b> | <b>72%</b> | <b>72%</b> | <b>72%</b> |

Table 3: Result and Analysis

# Result and Analysis

- The experimental results clearly demonstrate that the proposed Model outperforms all baseline architectures in terms of Precision, Recall, F1-score, and Accuracy see table 3.
- Compared to lightweight CNN models (table 2), the integration of advanced feature extraction and attention mechanisms significantly improves performance see .
- With an overall accuracy of 72%, the proposed approach provides a more reliable and efficient solution for practical deployment.

# Conclusions

This research proposes a lightweight deep learning model for cassava leaf disease classification that works effectively under real-world field conditions. The proposed model – which combines MobileNet2 as backbone with an attention mechanism and a leaf extraction pipeline – achieves 72% overall accuracy, outperforming other lightweight models like MobileNet (62%) and standard EfficientNet Lite (71%).

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# Thank You!